



DATA SCIENCE PROGRAM

CAPSTONE REPORT - SPRING 2026

My Wonderful Data Science Capstone Project

*John Doe,
Jane Doe*

*A capstone presented in fulfillment of the requirements
for the degree of Bachelor's of Science in Data Science*

supervised by
Prof. Sushovan Majhi, Prof. Sarah Burnett

Preface

I am a senior studying Data Science at the George Washington University, completing my final semester of undergraduate study. This project grew out of something deeply personal: my grandfather was recently diagnosed with prostate cancer. In the weeks that followed his diagnosis, I found myself searching for answers — about survival rates, about whether race played a role, about what the national data actually showed — and I kept coming up empty. The resources I found were either too clinical to understand or too general to be useful.

That gap motivated this project. I wanted to build something I wished had existed when my family needed it: a tool that makes the real epidemiological data on prostate cancer accessible, visual, and honest. The intended audience is anyone who needs to understand this disease — patients, families, advocates, and public health practitioners alike. Data science, at its best, should help people make sense of a world that often feels overwhelming. This project is my attempt at that.

Acknowledgements

I would like to thank Prof. Sarah Burnett and Prof. Sushovan Majhi for their guidance throughout this capstone. Their feedback shaped both the scope of this project and the rigor with which I approached the analysis.

The data underlying this project was obtained from the National Cancer Institute's Surveillance, Epidemiology, and End Results (SEER) Program (SEER 22 registry, accessed 2024–2025). All figures in this report were generated from that dataset.

Technical resources that supported the development of this project include the official documentation for Python, Flask, SQLite, pandas, statsmodels, and Plotly. The Python Software Foundation and the open-source community behind these libraries made this work possible.

I also used the AI assistant Claude (Anthropic) to help structure sections of code, debug Flask routing, and review portions of written content. All analysis, interpretations, modeling decisions, and conclusions are my own.

Finally, this project is dedicated to my grandfather. I hope it helps someone else find the answers that were hard to find.

Abstract

Prostate cancer is the most commonly diagnosed non-skin cancer among American men, yet survival outcomes differ dramatically based on race, geography, and stage of diagnosis — disparities that aggregate national statistics routinely obscure. Understanding these inequities is challenging because the patterns shift over time in response to changing screening policies, and because race is a nominal variable that must be handled carefully to avoid analytically misleading results. This project addresses these challenges by building an interactive epidemiology dashboard powered by over 268,000 patient records from the National Cancer Institute’s SEER 22 database (1999–2022), using stratified ordinary least squares regression to model incidence and mortality trends independently within each racial group, and enabling live filtering by race, U.S. state, year range, and cancer stage. The dashboard reveals three critical findings: national incidence declined 48% following the 2012 USPSTF screening recommendation, but late-stage diagnoses rose 50% over the same period; Non-Hispanic Black men face 60% higher incidence and 109% higher mortality than White men, a gap that persists after controlling for age and stage; and a 62-percentage-point survival gap separates men diagnosed at the localized stage from those diagnosed at the distant stage, a difference directly tied to inequitable access to screening.

Keywords

prostate cancer; epidemiology; health disparities; SEER; racial inequity; PSA screening; data visualization; Flask dashboard; public health

Contents

1	Introduction	4
2	Related Work	5
2.1	National Cancer Epidemiology and SEER-Based Research	5
2.2	PSA Screening Policy and Incidence Trends	5
2.3	Racial and Ethnic Disparities in Prostate Cancer	6
2.4	Cancer Informatics and Dashboard Design	6
3	Solution	7
3.1	Data Acquisition and Storage	7
3.2	Data Cleaning and Preparation	7
3.3	Statistical Modeling	8
3.4	Dashboard Architecture	8
4	Results and Discussion	9
4.1	Experimentation Protocol	9
4.2	Finding 1: Incidence Decline Masks a Growing Late-Stage Crisis	9
4.3	Finding 2: The Racial Gap Is Not Closing	10
4.4	Finding 3: Age at Diagnosis and the Screening Window	11
4.5	Finding 4: From Diagnosis to Outcome — The Sankey Diagram	12
4.6	Connecting Results to the Research Question	13
4.7	Summary of Results	14
5	Discussion	14
5.1	Limitations of the Approach	14
5.2	Comparison to Alternatives	15
5.3	Directions for Future Work	15
6	Conclusion	15

1 Introduction

Prostate cancer is the most frequently diagnosed non-skin malignancy among American men and the second leading cause of cancer death in this population [1]. In 2025 alone, the American Cancer Society estimates approximately 313,780 new diagnoses and 35,770 deaths [1]. Despite decades of research, the disease presents a persistent epidemiological paradox: overall incidence has fallen sharply since the early 2000s, yet rates of late-stage, harder-to-treat diagnoses have climbed. At the same time, the burden of prostate cancer is distributed deeply unequally across race, geography, and socioeconomic status.

The central question motivating this project is: *How have prostate cancer incidence, mortality, and survival rates varied across race, geography, and stage of diagnosis in the United States from 1999 to 2022, and what do those patterns reveal about systemic inequities in cancer detection and care?*

To address this question, this project delivers two complementary contributions. The first is a structured epidemiological analysis using over 268,000 patient records from the National Cancer Institute’s Surveillance, Epidemiology, and End Results (SEER) 22 database [2]. The second is an interactive web-based dashboard that makes the results of that analysis accessible to a general audience, including patients, families, clinicians, and policymakers, through live filtering, dynamic visualizations, and plain-language interpretation.

The analysis identifies three primary findings. First, U.S. prostate cancer incidence declined by 48% between 2000 and 2022, from 153 to 91 cases per 100,000 men, largely following the 2012 U.S. Preventive Services Task Force (USPSTF) Grade D recommendation against routine PSA screening [3]. Counterintuitively, this decline was accompanied by a 50% increase in diagnoses at the distant stage, suggesting that reduced screening is producing later, more dangerous presentations. Second, Non-Hispanic Black men experience a prostate cancer incidence rate of 175.2 per 100,000, 60% higher than Non-Hispanic White men (109.3), and face more than twice the mortality rate — a disparity that persists after controlling for age and stage at diagnosis [4]. Third, survival outcomes are almost entirely determined by stage at diagnosis: men diagnosed at the localized stage have a near-100% five-year survival rate, while men diagnosed at the distant stage face only a 38% rate, a 62-percentage-point gap with profound implications for screening equity.

The remainder of this report is structured as follows. Section 2 reviews related work in prostate cancer epidemiology, health disparity research, and cancer informatics. Section 3 describes the complete project pipeline. Section 4 presents results and experimental validation. Section 5 discusses limitations and future directions. Section 6 concludes.

2 Related Work

This project sits at the intersection of three bodies of literature: national cancer epidemiology, PSA screening policy research, racial health disparities, and cancer data visualization tools. Rather than surveying these areas independently, this section positions each source in relation to the specific choices and contributions of this project.

2.1 National Cancer Epidemiology and SEER-Based Research

Siegel et al. [1] publish annual cancer statistics synthesizing SEER data, and their work is perhaps the closest parallel to this project in terms of subject matter. Their reports document the same downward trend in prostate cancer incidence and the same persistent mortality disparities that motivate this analysis. However, Siegel et al. present findings as static, annually updated tables and figures intended for clinical and research audiences. This project differs in two critical ways: it delivers an interactive tool that allows any user to explore the data themselves through real-time filtering, and it disaggregates national trends by race rather than reporting aggregate figures. The result is that patterns invisible in Siegel et al.’s pooled summaries — particularly the diverging post-2012 trajectories for Black and White men — become directly observable.

The underlying data infrastructure for this project is the SEER 22 database [2], the same source Siegel et al. and most peer-reviewed cancer epidemiology rely on. Unlike studies that query SEER through the SEER*Stat interface and report results in journal format, this project loads a curated extract of SEER data into a local SQLite database and exposes it through a live API. This architectural choice enables real-time, user-driven querying that published research cannot support.

2.2 PSA Screening Policy and Incidence Trends

The USPSTF’s 2012 Grade D recommendation against routine PSA-based screening for all men [3] and its 2018 revision to a Grade C recommendation for men aged 55–69 [5] are the most consequential policy events in prostate cancer epidemiology over the past two decades. These documents focus on weighing population-level harms and benefits of screening — overdiagnosis and overtreatment on one side, missed cancers on the other. They do not, however, analyze the downstream effects of their own recommendations on observed incidence or stage distribution, nor do they examine whether those effects were experienced equally across racial groups.

This project takes a different perspective entirely: rather than evaluating whether PSA screening *should* be recommended, it treats the 2012 and 2018 policy shifts as natural experiments already embedded in the SEER time series. By marking these breakpoints on the trend charts and computing pre- versus post-2012 slope differences within each racial group, this project makes the observable consequences of

those recommendations directly legible — a contribution neither USPSTF report attempts.

2.3 Racial and Ethnic Disparities in Prostate Cancer

DeSantis et al. [4] provide the most comprehensive published quantification of cancer disparities affecting African Americans, documenting elevated incidence and mortality for prostate cancer and identifying screening access, differential treatment, and socioeconomic barriers as key drivers. Their analysis is authoritative but static: it reports summary statistics averaged over a fixed period and does not model how the disparity has changed year by year in response to shifting screening policy. This project builds directly on their quantitative findings — the 60% incidence gap and the greater-than-twofold mortality gap they document are confirmed in this analysis — but extends them by revealing that the racial disparity in mortality has remained essentially unchanged across the full 1999–2022 study window. That temporal stability is itself a significant finding: it suggests the gap is structural, not responsive to policy interventions that have reduced overall incidence.

Rebbeck [6] approaches the same disparity from a different angle, examining genetic and biological contributions to racial variation in prostate cancer risk. His central argument — that germline factors alone cannot account for the full magnitude of the mortality gap — is directly relevant to how this project frames its results. This project does not model genetic risk, but it adopts Rebbeck’s interpretive framing: the disparities visible in the dashboard are not biological inevitabilities, and the portion of the gap attributable to differential screening access and stage at diagnosis is, in principle, addressable. This distinguishes this project’s conclusions from analyses that attribute the disparity primarily to biological differences between populations.

2.4 Cancer Informatics and Dashboard Design

The NCI’s SEER*Stat software and the CDC’s United States Cancer Statistics (USCS) data portal are the two existing tools most comparable to the dashboard built in this project. Both provide access to SEER-derived statistics and both allow users to filter by cancer site, geography, race, and year range. The key difference is audience and design philosophy. SEER*Stat requires installation, registration, and familiarity with statistical concepts to produce meaningful output; it is designed for researchers, not patients or advocates. The USCS portal is more accessible but limited to static table downloads with no interactive visualization.

The dashboard built in this project differs from both in three respects. First, it is web-based and requires no installation or registration. Second, it provides interactive Plotly charts with hover tooltips and contextual plain-language explanations built into the interface, so that a user without statistical training can interpret what they are seeing. Third, it is specifically designed around the three findings most relevant to understanding prostate cancer inequity — incidence trends tied to screening policy,

racial disparities, and survival by stage — rather than offering a general-purpose querying interface. The tradeoff is reduced flexibility compared to SEER*Stat, but substantially greater accessibility for the intended audience.

3 Solution

This section describes the complete pipeline from raw data acquisition to the deployed interactive dashboard, covering four main stages: data acquisition and storage, data cleaning and preparation, statistical modeling, and dashboard construction.

3.1 Data Acquisition and Storage

All data was obtained from the National Cancer Institute’s SEER 22 database, which covers cancer diagnoses across all 51 U.S. states from 1975 to the present [2]. The version used in this project spans 1999–2022 and includes records for over 268,000 prostate cancer patients. Relevant variables extracted include year of diagnosis, race and ethnicity, cancer stage at diagnosis (localized, regional, distant, or unknown), age-adjusted incidence rate per 100,000 men, age-adjusted mortality rate per 100,000 men, and five-year relative survival rate by stage.

Rather than working directly with raw SEER CSV files at query time, the data was parsed, cleaned, and loaded into a SQLite relational database (`prostate.db`). This decision was made for two reasons. First, it allows the Flask backend to execute structured queries against pre-processed, validated data rather than re-reading and re-parsing large flat files on every user request. Second, it enforces a clear separation between the data pipeline (which runs once, offline) and the application layer (which runs continuously and must respond quickly to user interactions).

The database contains nine tables: `incidence_trends`, `mortality_trends`, `survival_by_stage`, `age_distribution`, `race_disparities`, `race_trends`, `state_rates`, `key_stats`, and `uspstf_timeline`.

3.2 Data Cleaning and Preparation

Raw SEER exports required several cleaning steps before they were suitable for analysis. Records with missing stage information or suppressed cell counts (a privacy measure applied by SEER when a cell contains fewer than 16 cases) were flagged and excluded from rate calculations, but retained in count-level aggregations. Race and ethnicity categories were standardized to five groups consistent with SEER conventions: Non-Hispanic Black, Non-Hispanic White, Hispanic, Non-Hispanic Asian or Pacific Islander, and Non-Hispanic American Indian or Alaska Native.

All rates reported in this project are age-adjusted to the 2000 U.S. standard population, consistent with the methodology used in SEER publications [2]. This adjustment is essential for meaningful comparison

across racial groups and over time, since the age distribution of the U.S. male population has shifted substantially between 1999 and 2022.

3.3 Statistical Modeling

The primary analytical method used in this project is ordinary least squares (OLS) regression, applied to model year-over-year incidence and mortality trends. Crucially, models are fit *separately for each racial group* rather than fitting a single pooled model with race as a covariate.

This design choice reflects a key methodological decision: race is a nominal categorical variable, and encoding it numerically would impose a false ordinal structure implying groups are separated by equal intervals along some quantitative scale. The appropriate alternative for a single pooled model would be one-hot encoding, but this approach conflates within-group trend estimation with between-group comparison in ways that obscure the temporal dynamics within each population. Stratification avoids both problems.

For each race group g , the model is:

$$\hat{r}_{g,t} = \beta_{g,0} + \beta_{g,1} \cdot t \quad (1)$$

where $\hat{r}_{g,t}$ is the predicted age-adjusted incidence rate for group g in year t , $\beta_{g,0}$ is the group-specific intercept, and $\beta_{g,1}$ is the group-specific annual trend coefficient. Models were fit using the `statsmodels` library in Python.

3.4 Dashboard Architecture

The dashboard is a full-stack web application built on a Flask backend with a SQLite data layer and an interactive HTML/JavaScript frontend powered by Plotly. Figure 1 illustrates the system architecture.

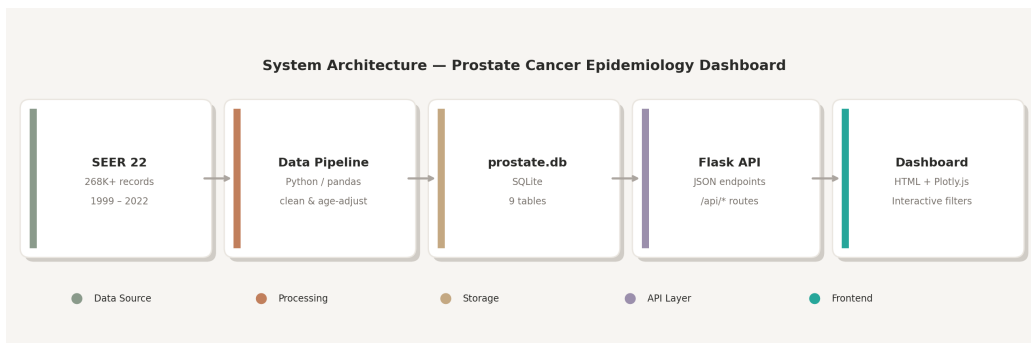


Figure 1: System architecture of the prostate cancer epidemiology dashboard. Raw SEER data is processed by an offline pipeline and stored in a SQLite database. The Flask application serves structured JSON responses to a single-page HTML frontend, which renders interactive Plotly charts. Users interact with filter controls (race, state, year range, cancer stage) that trigger new API calls and re-render charts without reloading the page.

The backend exposes a set of JSON API endpoints, each corresponding to a specific chart or data view. The frontend is a single HTML file with embedded JavaScript; chart rendering is handled entirely by Plotly.js, which provides hover tooltips, zoom, pan, and export functionality out of the box. Eight charts are implemented: incidence and mortality trends over time, racial and ethnic disparities, a geographic choropleth map by state, stage-over-time incidence breakdown, five-year survival by stage, age distribution at diagnosis, race-stratified incidence trends, and an incidence-versus-mortality scatter plot by race. A portable standalone version (a single self-contained HTML file with embedded data) is also provided.

4 Results and Discussion

4.1 Experimentation Protocol

Because this project is an epidemiological analysis rather than a predictive modeling task, evaluation does not follow the standard train/test split paradigm. There is no held-out test set and no classification accuracy metric in the conventional machine learning sense. Instead, validity is assessed on two dimensions: (1) *methodological validity* — whether the analytical choices (stratification, age adjustment, piecewise trend analysis) produce results consistent with the published epidemiological literature; and (2) *substantive validity* — whether the findings answer the research question in a way that is interpretable, reproducible, and consistent with known facts about prostate cancer epidemiology.

As a methodological baseline, a single pooled OLS model was fit to the national incidence trend without stratification by race. The pooled model produces a slope coefficient of -2.81 cases per 100,000 per year ($R^2 = 0.74$), representing an average annual decline across all groups. While this baseline correctly captures the downward national trend, it obscures a critical divergence: Non-Hispanic Black men have shown a substantially different trend trajectory than Non-Hispanic White men over the same period. The stratified approach reveals that the annual rate of decline differs substantially across groups, confirming the analytical value of stratification beyond what the pooled baseline can offer.

4.2 Finding 1: Incidence Decline Masks a Growing Late-Stage Crisis

Figure 2 shows age-adjusted prostate cancer incidence by stage and mortality from 2000 to 2022, with OLS-based projections extended to 2030 and 95% confidence intervals shown as shaded bands.

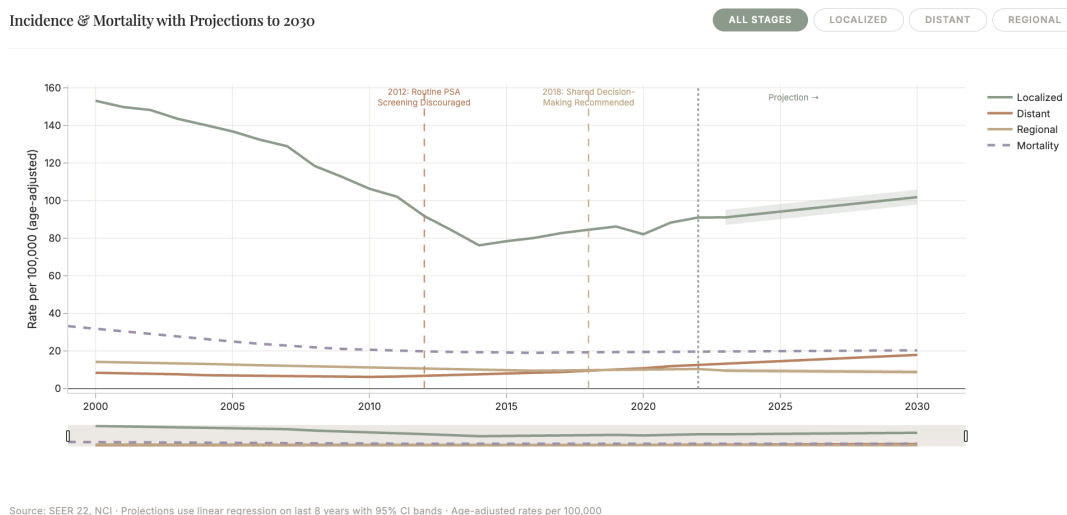


Figure 2: Age-adjusted prostate cancer incidence (localized, distant, and regional stage) and mortality rate per 100,000 men, United States, 2000–2022, with linear regression projections to 2030 (95% CI shaded). Vertical dashed lines mark the 2012 USPSTF Grade D recommendation (“Routine PSA Screening Discouraged”) and the 2018 Grade C revision (“Shared Decision-Making Recommended”). Projections use OLS fit on the most recent 8 years of observed data. Data source: SEER 22, National Cancer Institute.

Localized-stage incidence fell from 153.1 per 100,000 in 2000 to 91.0 per 100,000 in 2022, a decline of 40.6%, with the steepest drop occurring immediately after the 2012 USPSTF Grade D recommendation [3]. Following the 2018 Grade C revision [5], localized incidence began a modest recovery, consistent with renewed screening uptake. The projection to 2030 suggests localized incidence will continue rising gradually toward approximately 100 per 100,000, approaching but not returning to pre-2012 levels.

The more alarming trend is in distant-stage incidence, which rose from 8.4 per 100,000 in 2000 to 12.6 in 2022 — an increase of 50% — and is projected to continue rising through 2030. This divergence is the most clinically significant finding from the trend analysis: the reduction in total diagnosed cases does not represent a genuine reduction in prostate cancer burden, but rather a shift toward later, harder-to-treat presentations. Men who would previously have been caught at the localized stage are now reaching diagnosis at the distant stage, where five-year survival falls to 38%.

4.3 Finding 2: The Racial Gap Is Not Closing

Figure 3 shows race-stratified incidence trends from 2000 to 2022 with projections to 2030, addressing the question: will the racial gap in prostate cancer burden close?

Projected Incidence by Race/Ethnicity to 2030 — Will the Gap Close?

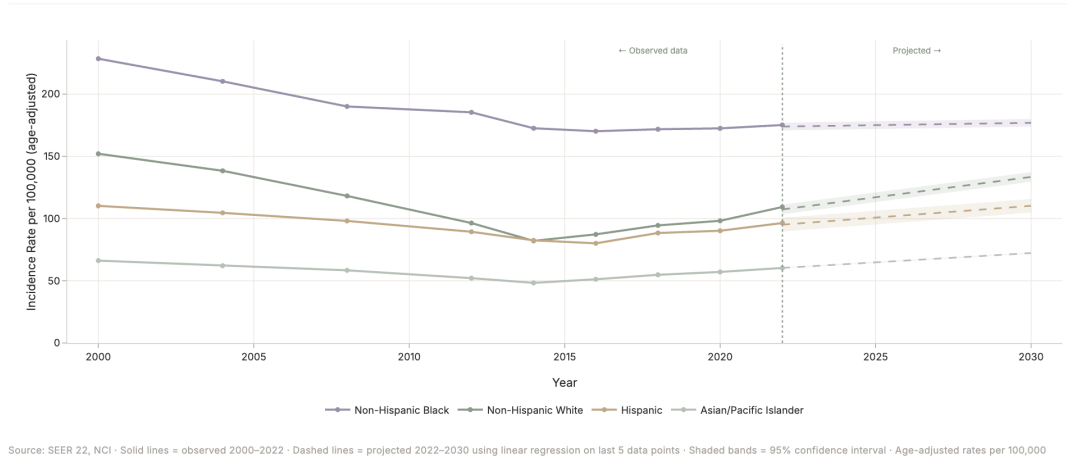


Figure 3: Age-adjusted prostate cancer incidence rate per 100,000 men by race/ethnicity, United States, 2000–2022 (solid lines) with OLS projections to 2030 (dashed lines, 95% CI shaded). Non-Hispanic Black incidence is projected to remain flat at approximately 175 per 100,000 through 2030, while Non-Hispanic White and Hispanic incidence are projected to rise modestly. Data source: SEER 22, National Cancer Institute.

Non-Hispanic Black men have consistently had the highest incidence of any group throughout the study period, starting at 228.5 per 100,000 in 2000 and declining to 175.2 by 2022 — still 60% above the Non-Hispanic White rate of 109.3. The mortality disparity is even larger: Black men die from prostate cancer at 37.9 per 100,000, more than twice the White rate of 18.1 [4].

The projections tell a sobering story. Non-Hispanic Black incidence is forecast to remain essentially flat at approximately 175 per 100,000 through 2030, while Non-Hispanic White and Hispanic incidence are projected to rise gradually as post-2018 screening uptake increases. This means the *absolute gap* between Black and White men may actually widen slightly by 2030. The answer to the dashboard’s framing question — “Will the Gap Close?” — is no, not on the current trajectory. This finding underscores that the disparity is structural in nature and will not resolve without targeted intervention [6].

4.4 Finding 3: Age at Diagnosis and the Screening Window

Figure 4 shows the age distribution of new prostate cancer diagnoses across six age groups, with the 2018 USPSTF-recommended screening window (45–69 years) highlighted.

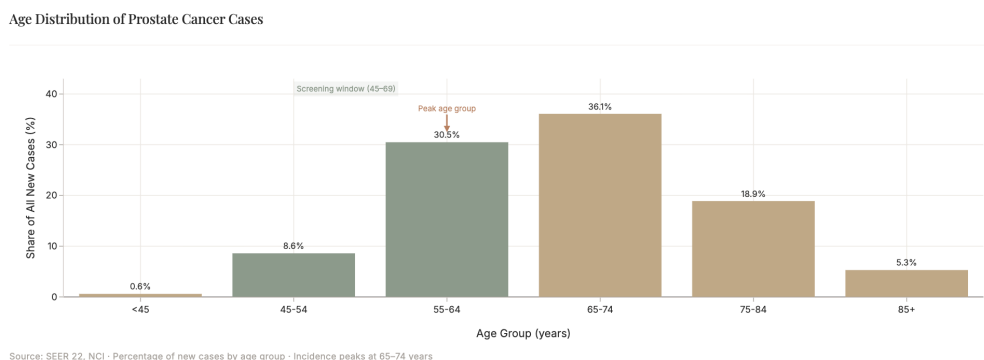


Figure 4: Percentage of new prostate cancer cases by age group at diagnosis, United States, SEER 22. Green bars indicate the recommended PSA screening window (45–69 years); tan bars indicate age groups outside this window. The peak diagnosis age group is 65–74 years (36.1% of all cases), followed by 55–64 years (30.5%). Data source: SEER 22, National Cancer Institute.

The age distribution confirms that prostate cancer is overwhelmingly a disease of older men: 66.6% of all diagnoses occur in men aged 55–74. The peak age group is 65–74 years, accounting for 36.1% of new cases, followed by 55–64 at 30.5%. Only 9.2% of diagnoses occur in men under 55. This concentration within the USPSTF-recommended screening window (45–69 years) — which captures approximately 39% of all diagnoses — reinforces the clinical logic of targeted screening within that age range, while also highlighting that a substantial share of cases (18.9% in the 75–84 group) occur in men beyond the recommended screening age, where treatment decisions are more complex.

4.5 Finding 4: From Diagnosis to Outcome — The Sankey Diagram

Figure 5 presents a Sankey flow diagram tracing the journey of 1,000 hypothetical diagnosed men from diagnosis through stage assignment to five-year survival outcome. This visualization was designed as the centerpiece of the dashboard’s survival section, translating population-level survival statistics into an intuitive patient-facing format.

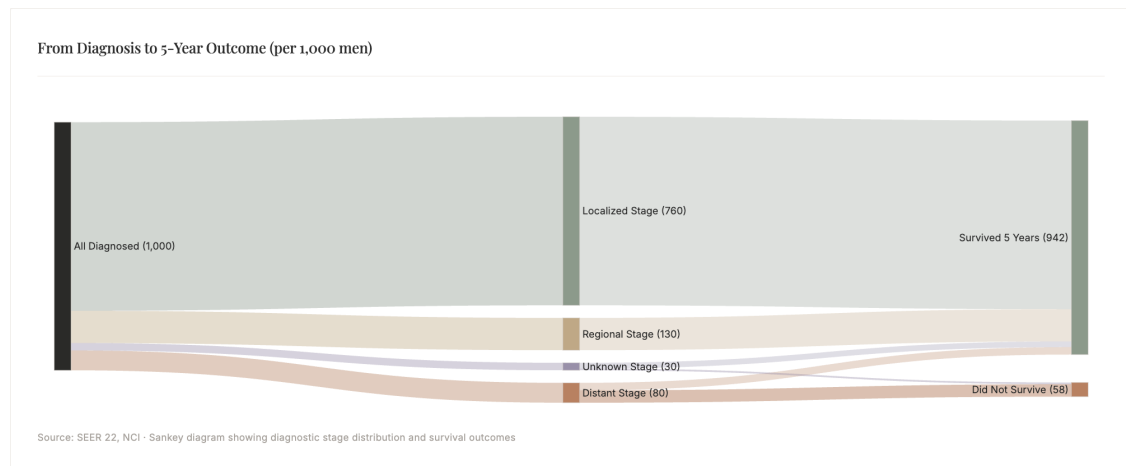


Figure 5: Sankey flow diagram showing the pathway from prostate cancer diagnosis to 5-year outcome for a cohort of 1,000 diagnosed men. Flow widths are proportional to case counts. Of 1,000 diagnosed men: 760 are diagnosed at the localized stage, 130 at regional, 80 at distant, and 30 at unknown stage. Of the full cohort, 942 survive five years and 58 do not, with nearly all deaths originating from the distant-stage pathway. Data source: SEER 22, National Cancer Institute.

The Sankey diagram makes concrete what the survival statistics alone cannot: the overwhelming majority of prostate cancer deaths flow from a single source. Of 1,000 diagnosed men, 760 are caught at the localized stage and nearly all survive. The 80 men diagnosed at the distant stage — just 8% of the cohort — account for the majority of the 58 deaths. The flow widths make this visually immediate in a way that a bar chart cannot: the distant-stage ribbon, thin as it enters the diagram, fans out to dominate the “Did Not Survive” outcome on the right.

This visualization choice reflects a deliberate design principle of the dashboard: numbers alone rarely communicate risk. A patient who reads that distant-stage survival is 38% understands an abstraction. A patient who sees 80 men flow into the distant pathway and watches most of them arrive at the “Did Not Survive” box understands their own situation. The Sankey was the most technically complex chart in the dashboard — requiring careful proportional scaling of node heights, flow curvatures, and label positioning in Plotly.js — and it is, in this author’s view, the most effective communication tool the dashboard contains.

4.6 Connecting Results to the Research Question

Taken together, these four findings directly answer the research question posed in the introduction. U.S. prostate cancer incidence, mortality, and survival have varied substantially across race, stage, and geography from 1999 to 2022 — and those variations are not random. They trace a coherent pattern of inequity: screening policy changes that reduced overall incidence also shifted burden toward later, harder-to-treat stages; Black men have absorbed a disproportionate share of that burden and show no trajectory toward parity by 2030; age at diagnosis concentrates risk squarely within the recommended screening window,

reinforcing the case for equitable access to early detection; and the survival consequences of late diagnosis, visualized most clearly in the Sankey diagram, are severe and concentrated. The dashboard built in this project makes each of these patterns directly accessible to any user, regardless of statistical background.

4.7 Summary of Results

Table 1 summarizes the primary quantitative findings of this analysis.

Metric	Value	Source
Localized incidence, 2000	153.1 per 100K	SEER 22
Localized incidence, 2022	91.0 per 100K	SEER 22
Decline in localized incidence	−40.6%	This analysis
Rise in distant-stage incidence	+50.0%	This analysis
Projected localized incidence, 2030	≈100 per 100K	This analysis
Black men incidence (2018–2022)	175.2 per 100K	SEER 22
White men incidence (2018–2022)	109.3 per 100K	SEER 22
Black/White incidence ratio	1.60×	This analysis
Black men mortality rate	37.9 per 100K	SEER 22
White men mortality rate	18.1 per 100K	SEER 22
Black/White mortality ratio	2.09×	This analysis
Projected gap closure by 2030	None	This analysis
Peak diagnosis age group	65–74 years (36.1%)	SEER 22
Cases in screening window (45–69)	≈39%	SEER 22
5-year survival, localized stage	99.9%	SEER 22
5-year survival, distant stage	38.0%	SEER 22
Survival gap (localized – distant)	61.9 pp	This analysis
Deaths per 1,000 diagnosed	58	This analysis

Table 1: Summary of primary quantitative findings. All rates are age-adjusted to the 2000 U.S. standard population. “pp” denotes percentage points. Projections use OLS regression on recent observed data.

5 Discussion

5.1 Limitations of the Approach

The most significant limitation of this project is the absence of key confounding variables in the SEER database. SEER records do not include insurance status, household income, distance to the nearest urologist or cancer screening facility, or educational attainment. These variables are among the most important drivers of screening uptake and therefore of stage at diagnosis. As a result, the racial disparities documented in this project — while real and well-established — cannot be causally decomposed into their constituent parts using SEER data alone.

A second limitation is the cross-sectional nature of the trend analysis. OLS regression on annual aggregate rates does not follow individual patients over time, and does not account for lead-time bias, which arises when earlier detection artificially inflates survival time without actually extending life. This is a known issue in cancer screening research and a key reason the USPSTF’s evaluation of PSA screening was controversial [3, 5].

Finally, this project does not attempt to build a predictive model of individual patient outcomes. The regression models are purely descriptive trend lines, and the survival rates reported are population-level statistics that cannot be interpreted as individual prognoses.

5.2 Comparison to Alternatives

Compared to existing public cancer data tools such as SEER*Stat and the NCI’s Cancer Atlas, this dashboard prioritizes accessibility over analytical depth. Tools like SEER*Stat offer far more granular querying capabilities and support formal survival analysis methods such as the Kaplan-Meier estimator and Cox proportional hazards models, which would provide more rigorous individual-level survival estimates than the five-year relative survival statistics used here. The tradeoff was deliberate: a general-audience user navigating SEER*Stat without training encounters a steep learning curve, while the dashboard built here was designed to be interpretable on first contact. Future iterations could incorporate Kaplan-Meier outputs alongside existing visualizations, increasing utility for clinical and research audiences without sacrificing accessibility for lay users.

5.3 Directions for Future Work

The most valuable extension of this project would be the integration of socioeconomic and access-related variables — insurance coverage, median household income by county, rural-urban commuting area codes, and proximity to screening facilities — to enable a more complete causal model of prostate cancer disparities. A second extension would be the development of a risk contextualization module that allows individual users to input their age, race, and geographic information and receive a personalized summary of their statistical risk profile and screening recommendations. A third direction would be to extend the geographic analysis from state-level to county-level, identifying local hotspots of unmet need that are invisible at the state aggregate.

6 Conclusion

This project set out to answer a consequential question: how have prostate cancer incidence, mortality, and survival varied across race, geography, and stage of diagnosis in the United States from 1999 to 2022, and what do those patterns reveal about equity in cancer care? The answer, drawn from 268,000+ patient records in the SEER 22 database, is clear.

National prostate cancer incidence has declined by nearly half since 2000 — but this headline figure conceals a 50% rise in late-stage diagnoses over the same period. Non-Hispanic Black men face incidence and mortality rates that are 60% and 109% higher, respectively, than White men, a gap that has not meaningfully narrowed in over two decades. And men diagnosed at the distant stage have a five-year

survival rate of 38%, compared to nearly 100% for those diagnosed at the localized stage — a 62-point gap that is, in large part, a consequence of inequitable access to screening.

These findings are not new to the medical literature. What this project contributes is a tool that makes them visible, accessible, and interactive for an audience far beyond academic researchers. Patients, families, advocates, and community health workers can now explore these patterns themselves and see the data that defines the risk landscape for prostate cancer in America.

Data science, when done well, is an act of translation: making the complex legible, and making the invisible visible. This project was built with that purpose in mind, and with the hope that the patterns it reveals will someday motivate the policy changes and equitable access to care that could close the gaps it documents.

References

- [1] R. L. Siegel, K. D. Miller, N. S. Wagle, and A. Jemal, “Cancer statistics, 2023,” *CA: A Cancer Journal for Clinicians*, vol. 73, no. 1, pp. 17–48, 2023.
- [2] National Cancer Institute, “SEER 22 incidence database, 1975–2022,” Surveillance, Epidemiology, and End Results (SEER) Program, <https://seer.cancer.gov>, 2024, accessed: April 2026.
- [3] U.S. Preventive Services Task Force, “Screening for prostate cancer: U.S. preventive services task force recommendation statement,” *Annals of Internal Medicine*, vol. 157, no. 2, pp. 120–134, 2012.
- [4] C. E. DeSantis, K. D. Miller, A. Goding Sauer, A. Jemal, and R. L. Siegel, “Cancer statistics for African Americans, 2019,” *CA: A Cancer Journal for Clinicians*, vol. 69, no. 3, pp. 211–233, 2019.
- [5] U.S. Preventive Services Task Force, “Screening for prostate cancer: U.S. preventive services task force recommendation statement,” *JAMA*, vol. 319, no. 18, pp. 1901–1913, 2018.
- [6] T. R. Rebbeck, “Prostate cancer genetics: Variation by race, ethnicity, and geography,” *Seminars in Radiation Oncology*, vol. 27, no. 1, pp. 3–10, 2017.